

Greedy expansions

This document is about summarizing everything I know about greedy expansions, with the aim of exporting them easily to papers/appendices. We do everything for the general case $\beta \in (1, \infty)$.

In all what follows, $\mathbb{N}^{\mathbb{N}}$ and its subsets are endowed with the lexicographical ordering, $\mathbb{R}$ and its subsets are endowed with the natural ordering. $\{\cdot\}$ is the fractional part.

1 Definition and first properties

1.1 Rényi's greedy expansion

Historically, the *greedy β -expansion* was introduced by A. Rényi in [5, Section 4], under the simpler name of β -expansion. The definition relies on a map that, following [1, Abstract], we call β -transformation.

Definition 1.1. (*β -transformation*) Let $\beta \in (1, \infty)$. We define the β -transformation $T_\beta : [0, 1] \rightarrow [0, 1]$ by

$$T_\beta(s) = \{\beta s\}, \quad \forall s \in [0, 1]. \quad (1)$$

We also define the β -digitization $d_\beta : [0, 1] \rightarrow \mathbb{N}$ by

$$d_\beta(s) = \lfloor \beta s \rfloor, \quad \forall s \in [0, 1]. \quad (2)$$

Note that $T_\beta(s) = \beta s - d_\beta(s)$, $\forall s \in [0, 1]$.

Rényi defined the greedy β -expansion to be the successive digits generated from iterations of the β -transformations.

Definition 1.2. (*Rényi's greedy β -expansion*) Let $\beta \in (1, \infty)$. Then, we define the Rényi's greedy β -expansion $\mathbf{g}_\beta : [0, 1] \rightarrow \mathbb{N}^{\mathbb{N}}$ by

$$\mathbf{g}_\beta(s) = \prod_{i=0}^{\infty} d_\beta(T_\beta^i(s)), \quad \forall s \in [0, 1]. \quad (3)$$

For $s \in [0, 1]$, we say that $\mathbf{g}_\beta(s)$ is the greedy β -expansion of s .

The name greedy β -expansion of s is justified because $\mathbf{g}_\beta(s)$ is exactly analogous to the usual definition of the expansion of s in some integer base, as the next lemma shows.

Lemma 1.1. [5, Section 1] (*Rényi's greedy expansion represents real numbers*) Let $\beta \in (1, \infty)$, and $s \in [0, 1]$. Then,

$$s = \sum_{i=1}^{\infty} \mathbf{g}_\beta(s)_i \beta^{-i}. \quad (4)$$

Lemma 1.1 is infact a direct corollary of the following stronger statement, that quantifies how fast the Rényi's greedy β -expansion approximates the real numbers.

Lemma 1.2. (*Approximation rate of the Rényi's greedy expansion*) Let $\beta \in (1, \infty)$, and $s \in [0, 1]$. Then,

$$s = \sum_{i=1}^n \mathbf{g}_\beta(s)_i \beta^{-i} + T_\beta^n(s) \beta^{-n}, \quad \forall n \in \mathbb{N}. \quad (5)$$

Concretely, the partial sum $\sum_{i=1}^n \mathbf{g}_\beta(s)_i \beta^{-i}$ approximates s with error $T_\beta^n(s) \beta^{-n}$.

Proof. Let $\beta \in (1, \infty)$, $s \in [0, 1]$ and $n \in \mathbb{N}$. Note that

$$\frac{1}{\beta^n} T_\beta^n(s) = \frac{1}{\beta^n} T_\beta(T_\beta^{n-1}(s)) = \frac{1}{\beta^n} \{\beta T_\beta^{n-1}(s)\} \quad (6)$$

$$= \frac{1}{\beta^n} \beta T_\beta^{n-1}(s) - \frac{1}{\beta^n} d_\beta(T_\beta^{n-1}(s)) \quad (7)$$

$$= \frac{1}{\beta^{n-1}} T_\beta^{n-1}(s) - \frac{1}{\beta^n} \mathbf{g}_\beta(s)_n. \quad (8)$$

By induction on n , we get

$$\frac{1}{\beta^n} T_\beta^n(s) = \frac{1}{\beta^0} T_\beta^0(s) - \sum_{i=1}^n \frac{1}{\beta^i} \mathbf{g}_\beta(s)_i = s - \sum_{i=1}^n \frac{1}{\beta^i} \mathbf{g}_\beta(s)_i. \quad (9)$$

□

1.2 Erdős's greedy expansion

33 years later, P. Erdős defined in [3] the greedy β -expansion to be the lexicographically largest β -expansion, with the definition of β -expansion being quite different from the one of A. Rényi.

Definition 1.3 (β -expansion). *Let $\beta \in (1, \infty)$. We say that $\mathbf{x} \in \mathbb{N}^{\mathbb{N}}$ is a β -expansion of $s \in [0, 1]$ if*

$$s = \sum_{i=1}^{\infty} \mathbf{x}_i \beta^{-i} \text{ and } \mathbf{x}_i \in \mathbb{N}[\beta] - 1 \ \forall i \in \mathbb{N}. \quad (10)$$

We can reformulate the above equation. Let $\delta_\beta : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{R}$ be such that $\Omega_\beta := \text{dom}(\delta_\beta) := \{0, \dots, [\beta] - 1\}^{\mathbb{N}}$ and

$$\delta_\beta(\mathbf{x}) = \sum_{i=1}^{\infty} \mathbf{x}_i \beta^{-i}, \ \forall \mathbf{x} \in \Omega_\beta. \quad (11)$$

Then, Equation (10) can be rewritten

$$s = \delta_\beta(\mathbf{x}). \quad (12)$$

For $s \in [0, 1]$, we denote by $\Omega_\beta(s) \subset \Omega_\beta$ the set of all the β -expansions of s .

From this definition, we build-up to define the greedy β -expansion according to Erdős.

Definition 1.4. (*Erdős's greedy β -expansion*) *Let $\beta \in (1, \infty)$, and $s \in [0, 1]$. We say that $\sup \Omega_\beta(s)$ is the Erdős's greedy β -expansion of s .*

We can prove that actually, both definitions agree, as remarked originally by P. Erdős and collaborators, because they use the two definitions interchangeably in [2] and [3].

Theorem 1.3. (*Greedy β -expansion*)

$$\mathbf{g}_\beta(s) = \sup \Sigma_\beta(s), \ \forall s \in [0, 1]. \quad (13)$$

From now on, we simply call $\mathbf{g}_\beta$ the greedy β -expansion, and for $s \in [0, 1]$, we say that $\mathbf{g}_\beta(s)$ is the greedy β -expansion.

Proof of Theorem 1.3. Let $\beta \in (1, \infty)$ and $s \in [0, 1]$. First, Lemma 1.1 asserts that $\mathbf{g}_\beta(s) \in \Omega_\beta(s)$. Suppose that $\mathbf{g}_\beta(s) \neq \sup \Omega_\beta(s)$. Then, there exists $\mathbf{x} \in \Omega_\beta(s)$ such that $\mathbf{x} >_L \mathbf{g}_\beta(s)$, i.e., there exists $i \in \mathbb{N}$ such that $\mathbf{x}_j = \mathbf{g}_\beta(s)_j$ for all $j < i$, and $\mathbf{x}_i > \mathbf{g}_\beta(s)_i =: k$. Note that $k = d_\beta(T_\beta^{i-1}(s)) = \lfloor \beta T_\beta^{i-1}(s) \rfloor$, i.e. $T_\beta^{i-1}(s) \in [k/\beta, (k+1)/\beta)$. By Equation (5), one gets

$$(k+1)\beta^{-i} > T_\beta^{i-1}(s)\beta^{-i+1} = s - \delta_\beta(\mathbf{g}_\beta(s)|_{i-1}) = s - \delta_\beta(\mathbf{x}|_{i-1}). \quad (14)$$

Then,

$$\delta_\beta(\mathbf{x}) \geq \delta_\beta(\mathbf{x}|_i) = \delta_\beta(\mathbf{x}_{i-1}) + \beta^{-i} \mathbf{x}_i \geq \delta_\beta(\mathbf{x}_{i-1}) + \beta^{-i}(k+1) \stackrel{(14)}{>} s, \quad (15)$$

i.e., $\mathbf{x} \notin \Omega_\beta(s)$, which is a contradiction. $\square$

1.3 Properties of the set of greedy expansions

We first introduce Σ_β to be the set of greedy β -expansions, usually called β -shift [6], which in mathematical symbols is expressed as

$$\Sigma_\beta := \text{range}(\mathbf{g}_\beta) = \{\mathbf{g}_\beta(s) : s \in [0, 1]\} \subseteq \Omega_\beta. \quad (16)$$

From now on, we will study the behavior of $\mathbf{g}_\beta$ considered as a function from $[0, 1]$ to Σ_β . A first evident property is for $\mathbf{g}_\beta$ to be bijective.

Definition 1.5. (*Bijectivity*) $\mathbf{g}_\beta : [0, 1] \rightarrow \Sigma_\beta$ is bijective, of inverse $\delta_\beta|_{\Sigma_\beta}$.

Proof. By Lemma 1.1, $\mathbf{g}_\beta$ is injective and

$$\delta_\beta(\mathbf{g}_\beta(s)) = s, \ \forall s \in [0, 1]. \quad (17)$$

This makes $\mathbf{g}_\beta$ bijective onto its range, with inverse $\delta_\beta|_{\Sigma_\beta}$. $\square$

A following very important property is that $\mathbf{g}_\beta : ([0, 1], <) \rightarrow (\Sigma_\beta, <_L)$ is an order isomorphism, i.e. a order-preserving bijection with order-preserving inverse.

Lemma 1.4. (*Greedy expansion is an order isomorphism*) [4, Lemma 3] Let $s, t \in [0, 1]$, and $\beta \in (1, \infty)$. Then,

$$s < t \iff \mathbf{g}_\beta(s) <_L \mathbf{g}_\beta(t), \quad (18)$$

i.e. both $\mathbf{g}_\beta$ and $\delta_\beta|_{\Sigma_\beta}$ are order-preserving.

Proof. Let $s, t \in [0, 1]$, and $\beta \in (1, \infty)$. First note that, by bijectivity of $\mathbf{g}_\beta$ (Lemma 1.5), $s \neq t \iff \mathbf{g}_\beta(s) \neq \mathbf{g}_\beta(t)$. It then suffices to prove that (a) $s < t \Rightarrow \mathbf{g}_\beta(s) \leq_L \mathbf{g}_\beta(t)$ and (b) $\mathbf{g}_\beta(s) <_L \mathbf{g}_\beta(t) \Rightarrow s \leq t$. Note that (b) is simply the contraposition of (a), so we only need to prove (a). We proceed by contradiction: suppose that $s < t$, and that $\mathbf{g}_\beta(s) >_L \mathbf{g}_\beta(t)$, i.e., that there exists $i \in \mathbb{N}$ such that $\mathbf{g}_\beta(s)_j = \mathbf{g}_\beta(t)_j$, for all $j < i$, and $\mathbf{g}_\beta(s)_i > \mathbf{g}_\beta(t)_i$. Hence,

$$t > s \geq \sum_{j=1}^i \mathbf{g}_\beta(s)_j \beta^{-j} > \sum_{j=1}^i \mathbf{g}_\beta(t)_j \beta^{-j} \geq t - \beta^{-i}. \quad (19)$$

So $\mathbf{g}_\beta(s)_{1:i}$ is the i -prefix of some β -expansion $\mathbf{x}$ of t , satisfying $\mathbf{x} >_L \mathbf{g}_\beta(t)$, which is a contradiction. $\square$

We further study Σ_β by studying the properties of the finite prefixes of its elements. For $n \in \mathbb{N}$, we study the set

$$\Sigma_\beta(n) := \{\mathbf{x}|_n : \mathbf{x} \in \Sigma_\beta\} \quad (20)$$

of n -prefixes of all the greedy β -expansions. These n -prefixes constitute an approximation of the full infinite expansion, which motivates us to call $\Sigma_\beta(n)$ the set of approximate greedy β -expansion of order n . Note that $\Sigma_\beta(n)$ is a finite set of cardinality $N_\beta(n) := \#(\Sigma_\beta(n))$ less than 2^n . We denote each element of $\Sigma_\beta(n)$ by $x_i(n)$, for $i \in \{0, \dots, N_\beta(n)\}$, such that $x_0(n) <_L x_1(n) <_L \dots <_L x_{N_\beta(n)-1}(n)$. We also introduce the set of all finite prefixes of the greedy β -expansions as

$$\Sigma_\beta^* := \bigcup_{n=0}^{\infty} \Sigma_\beta(n). \quad (21)$$

At this point, it is handy to extend δ_β to finite sequences, through the following convention:

$$\delta_\beta(x) := \delta_\beta(x0^\infty), \quad \forall x \in \{0, \dots, \lceil \beta \rceil - 1\}^*. \quad (22)$$

The main first observation is that each element of $\Sigma_\beta(n)$ describes fully one real number.

Lemma 1.5. (*Greedy expansion is stable by appending 0*) Let $n \in \mathbb{N}$, and $\beta \in (1, \infty)$. Then,

$$\mathbf{g}_\beta(\delta_\beta(x)) = x0^\infty, \quad \forall x \in \Sigma_\beta(n). \quad (23)$$

Proof. Let $n \in \mathbb{N}$, $\beta \in (1, \infty)$ and $x \in \Sigma_\beta(n)$. By means of contradiction, suppose that $x0^\infty$ is not the greedy β -expansion of $s := \delta_\beta(x)$, which by Theorem 1.3 yields $\mathbf{y} := \mathbf{g}_\beta(\delta_\beta(x)) >_L x0^\infty =: \mathbf{x}$, i.e., there exists $i \in \mathbb{N}$ such that $\mathbf{y}_j = \mathbf{x}_j$, for all $j < i$, and $1 = \mathbf{y}_i > \mathbf{x}_i = 0$.

(a) Suppose that $i > n$. Then,

$$\delta_\beta(\mathbf{y}) \leq \delta_\beta(\mathbf{y}_{1:n}) + \beta^{-i} \mathbf{y}_i \quad (24)$$

$$= \delta_\beta(x) + \beta^{-i} \mathbf{y}_i \quad (25)$$

$$> \delta_\beta(x) = s, \quad (26)$$

so $\mathbf{y}$ is not a β -expansion of $\delta_\beta(x)$, which is a contradiction to $\text{eval}(\mathbf{y}) = s$.

(b) Suppose that $i \leq n$. Since $x \in \mathbf{g}_\beta([0, 1]_{1:n})$, there exists $s \in [0, 1]$ such that $\mathbf{g}_\beta(s)_{1:n} = x$. Note that

$$s = \delta_\beta(\mathbf{g}_\beta(s)) \geq \delta_\beta(\mathbf{g}_\beta(s)_{1:n}) = \delta_\beta(x), \quad (27)$$

and that

$$\mathbf{g}_\beta(s)_{1:i-1} = x_{1:i-1} = \mathbf{y}_{1:i-1} = \mathbf{g}_\beta(\delta_\beta(x))_{1:i-1}. \quad (28)$$

Hence,

$$T_\beta^{i-1}(s) = \beta^{i-1}(s - \delta_\beta(\mathbf{g}_\beta(s)_{i-1})) \quad (29)$$

$$\geq \beta^{i-1}(\delta_\beta(x) - \delta_\beta(\mathbf{g}_\beta(s)_{i-1})) \quad (30)$$

$$= \beta^{i-1}(\delta_\beta(x) - \delta_\beta(\mathbf{g}_\beta(\delta_\beta(x))_{i-1})) \quad (31)$$

$$= T_\beta^{i-1}(\delta_\beta(x)). \quad (32)$$

However, $\mathbf{g}_\beta(\delta_\beta(x))_i = \mathbf{y}_i = 1$, so $T_\beta^{i-1}(\delta_\beta(x)) > 1/\beta$, and $\mathbf{g}_\beta(s)_i = x_i = 0$, so $T_\beta^{i-1}(s) \leq 1/\beta$, which yields a contradiction.

□

We can immediately deduce a useful relation between $\Sigma_\beta(n)$ and $\Sigma_\beta(n+1)$, mentioned first in the proof of [5, Theorem 2].

Corollary 1.6. (*Approximation set stable by adding 0*)

$$x \in \Sigma_\beta(n) \Rightarrow x0 \in \Sigma_\beta(n+1). \quad (33)$$

Moreover, Lemma 1.5 allows establish a natural correspondance between $\Sigma_\beta(n)$ and the elements of the set of greedy β -adic numbers of order n $X_\beta(n) := \delta_\beta(\Sigma_\beta(n))$. Indeed, $\delta_\beta : \Sigma_\beta(n) \rightarrow X_\beta(n)$ is a bijection with inverse $\mathbf{g}_\beta|_n$. Furthermore, Lemma 1.4 shows that the elements $\Sigma_\beta(n)$ and $X_\beta(n)$ have the same order, with respect to $<_L$ and $<$, respectively. Hence, we denote the elements of $X_\beta(n) \cup \{1\}$ by s_i , for $i \in \{1, \dots, N_\beta(n)\}$, such that $0 = s_0(n) < s_1(n) < \dots < s_{N_\beta(n)-1}(n) \leq s_{N_\beta(n)}(n) = 1$, which by Lemma 1.4 yields $s_i(n) = \delta_\beta(x_i)$, for $i \leq N_\beta(n) - 1$. We also introduce the set of greedy β -adic numbers

$$X_\beta^* := \delta_\beta(\Sigma_\beta^*) = \bigcup_{n=0}^{\infty} X_\beta(n). \quad (34)$$

Note that since Σ_β is the closure of $\Sigma_\beta^* 0^\infty$ in the Baire space topology, then $[0, 1]$ is the closure of X_β^* in the Euclidean topology. The next result expresses the fact that we can define greedy β -expansion by considering the elements of X_β^* .

Lemma 1.7. *Let $\beta \in (1, \infty)$ and $s \in [0, 1]$. Then, for $n \in \mathbb{N}$,*

$$\delta_\beta(\mathbf{g}_\beta(s)|_n) = \sup\{s_i(n) \in X_\beta(n) : s_i(n) \leq s\}, \quad (35)$$

which also translates to

$$\mathbf{g}_\beta(s)|_n = \sup_L \{x_i(n) \in \Sigma_\beta(n) : s_i(n) \leq s\}. \quad (36)$$

Proof. Let $\beta \in (1, \infty)$, $s \in [0, 1]$ and $n \in \mathbb{N}$. First, obviously, $\delta_\beta(\mathbf{g}_\beta(s)|_n) \leq s$. Now, suppose that $s_j(n) := \delta_\beta(\mathbf{g}_\beta(s)|_n) < \sup\{s_i(n) \in X_\beta(n) : s_i(n) \leq s\} =: s_I(n)$. Then, by Lemma 1.4, $x_j(n) = \mathbf{g}_\beta(s_j(n))|_n <_L \mathbf{g}_\beta(s_I(n))|_n = x_I(n)$. Note that since $s_I(n) \in [s_j(n), s] \subseteq [s - \frac{[\beta]-1}{\beta-1}\beta^{-n}, s]$, $x_I(n)$ is the n -prefix of β -expansion of s . However, since $x_I(n) >_L x_j(n)$, we have a contradiction to the fact that $x_j(n)$ is the a prefix of the greedy β -expansion of s . □

We first derive an upper bound on the spacings between the consecutive elements of $X_\beta(n)$.

Lemma 1.8. (*Spacings between elements of $X_\beta(n)$*) *Let $\beta \in (1, \infty)$, and $n \in \mathbb{N}$. Then, for every $i \in \{0, \dots, N_\beta(n) - 1\}$, $s_{i+1}(n) \leq s_i(n) + \beta^{-n}$.*

Proof. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $r \in [s_i(n), s_{i+1}(n))$. By Lemma 1.1, we have that

$$r = \delta_\beta(\mathbf{g}_\beta(r)|_n) + T_\beta^n(r)\beta^{-n} \stackrel{\text{Lem. 1.7}}{=} s_i(n) + T_\beta^n(r)\beta^{-n} < s_i(n) + \beta^{-n}. \quad (37)$$

Hence,

$$s_{i+1}(n) = \lim_{r \rightarrow s_{i+1}(n)} r \leq \lim_{r \rightarrow s_{i+1}(n)} s_i(n) + \beta^{-n} = s_i(n) + \beta^{-n}. \quad (38)$$

□

A fundamental question about the nature of $\Sigma_\beta(n)$ (or equivalently $X_\beta(n)$) is its cardinality. The behavior of $N_\beta(n)$ was first established in the proof of [5, Theorem 2]. Note that by Lemma 1.6, we get that $X_\beta(n-1) \subseteq X_\beta(n)$. It will prove useful to consider closely the real numbers in $X_\beta(n)$ that lie between two consecutive numbers in $X_\beta(n-1)$. Accordingly, we define

$$X_\beta(n, i) := X_\beta(n) \cap [s_i(n-1), s_{i+1}(n-1)), \quad (39)$$

and

$$\Sigma_\beta(n, i) := \mathbf{g}_\beta(X_\beta(n, i))|_n, \quad (40)$$

for all $i \in \{0, \dots, N_\beta(n) - 1\}$.

Lemma 1.9. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $i \in \{0, \dots, N_\beta(n) - 1\}$. Then,

$$\Sigma_\beta(n, i) \subseteq x_i(n-1) \cdot \{0, \dots, \lceil \beta \rceil - 1\}. \quad (41)$$

Proof. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $i \in \{0, \dots, N_\beta(n) - 1\}$, and suppose that

$$\Sigma_\beta(n, i) \not\subseteq x_i(n-1) \cdot \{0, \dots, \lceil \beta \rceil - 1\}. \quad (42)$$

Then, there exists $x \in \{0, \dots, \lceil \beta \rceil - 1\}^{n-1}$ and $b \in \{0, \dots, \lceil \beta \rceil - 1\}$, such that $x \neq x_i(n-1)$ and $xb \in \Sigma_\beta(n, i)$. Hence, $\delta_\beta(xb) \in X_\beta(n, i)$, i.e. $s_i(n-1) = \sup\{s_j(n-1) \in X_\beta(n-1) : s_j(n-1) \leq \delta_\beta(xb)\}$, which implies that

$$x \stackrel{\text{Lem. 1.5}}{=} \mathbf{g}_\beta(\delta_\beta(xb))|_{n-1} \stackrel{\text{Lem. 1.7}}{=} \mathbf{g}_\beta(s_i(n-1))|_{n-1} = x_i(n-1), \quad (43)$$

which yields a contradiction. $\square$

Lemma 1.10. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $i \in \{0, \dots, N_\beta(n) - 1\}$. Then, there exists $k \in \{0, \dots, \lceil \beta \rceil - 1\}$ such that

$$\Sigma_\beta(n, i) = x_i(n-1) \cdot \{0, \dots, k\}. \quad (44)$$

Proof. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $i \in \{0, \dots, N_\beta(n) - 1\}$. First, by previous Lemma, we have that $\Sigma_\beta(n, i) \subseteq x_i(n-1) \cdot \{0, \dots, \lceil \beta \rceil - 1\}$. Let $j > 0$ be such that $x_i(n-1)j \in \Sigma_\beta(n, i)$. Suppose that $x_i(n-1)j - 1 \notin \Sigma_\beta(n, i)$. Then, $\square$

Lemma 1.11. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$, and $i \in \{0, \dots, N_\beta(n) - 1\}$. Then,

$$|\text{cvxhull}(X_\beta(n, i))| \geq \beta^{-n}(\#X_\beta(n, i) - 1). \quad (45)$$

Proof. $\square$

Lemma 1.12. Let $\beta \in (1, \infty)$, $n \in \mathbb{N}$. Then,

$$N_\beta(n) \leq \frac{\beta^{n+1}}{\beta - 1}. \quad (46)$$

Proof.

$$N_\beta(n) = \#X_\beta(n) = \sum_{i=0}^{N_\beta(n-1)-1} \#X_\beta(n, i) \quad (47)$$

$$\leq \sum_{i=0}^{N_\beta(n-1)-1} (\beta^n |\text{cvxhull}(X_\beta(n, i))| + 1) \quad (48)$$

$$= \beta^n \sum_{i=0}^{N_\beta(n-1)-1} |\text{cvxhull}(X_\beta(n, i))| + N_\beta(n-1) \quad (49)$$

$$\leq \beta^n + N_\beta(n-1). \quad (50)$$

Hence, since $N_\beta(0) = 1$, we get that

$$N_\beta(n) \leq \sum_{i=1}^n \beta^i = \frac{\beta^{n+1} - 1}{\beta - 1} \leq \frac{\beta^{n+1}}{\beta - 1}. \quad (51)$$

$\square$

Theorem 1.13. (Growth rate of $\#(\Sigma_\beta(n))$) Let $\beta \in (1, \infty)$, and $n \in \mathbb{N}$. Then,

$$\beta^n \leq N_\beta(n) \leq \frac{\beta^{n+1}}{\beta - 1}. \quad (52)$$

Proof. Let $\beta \in (1, \infty)$, and $n \in \mathbb{N}$. The left inequality follows from Lemma 1.8, as

$$1 = s_{N_\beta(n)} = \sum_{i=0}^{N_\beta(n)-1} (s_{i+1} - s_i(n)) + s_0(n) = \sum_{i=0}^{N_\beta(n)-1} (s_{i+1} - s_i(n)) \leq N_\beta(n)\beta^{-n}, \quad (53)$$

which yields $N_\beta(n) \geq \beta^n$. We move to the right inequality. $\square$

References

- [1] K. Dajani, C. Kraaikamp, and B. Solomyak, “The natural extension of the β -transformation,” 1995.
- [2] P. Erdős and M. Horváth, “On the uniqueness of expansions $1 = \sum_{i=1}^{\infty} q^{-n_i}$,” *Acta Mathematica Hungarica*, 1991.
- [3] P. Erdős, V. Komornik, and I. Joó, “Characterization of the unique expansions $1 = \sum_{i=1}^{\infty} q^{-n_i}$ and related problems,” *Bulletin de la Société Mathématique de France*, 1990.
- [4] W. Parry, “On the β -expansions of real numbers,” *Acta Mathematica Hungarica*, vol. 11, no. 3-4, pp. 401–416, 1960.
- [5] A. Rényi, “Representations for real numbers and their ergodic properties,” *Acta Mathematica Academiae Scientiarum Hungarica*, vol. 8, no. 3-4, pp. 477–493, 1957.
- [6] J. Schmerling, “Symbolic dynamics for beta-shifts and self-normal numbers,” *Ergodic Theory and Dynamical Systems*, vol. 17, pp. 675–694, 1995.